1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
D3_96509_Aditya> select o.onum,o.amt,o.odate from orders o WHERE o.cnum =
    -> (select c.cnum  FROM customers c WHERE c.cname = 'Cisneros');
+------+---------+------------+
| onum | amt     | odate      |
+------+---------+------------+
| 3001 |   18.69 | 1990-10-03 |
| 3006 | 1098.16 | 1990-10-03 |
+------+---------+------------+
2 rows in set (0.00 sec)
```

7. Extract all the orders of Motika.

```
D3_96509_Aditya> select * from orders o WHERE o.snum =
    ->  (select snum FROM salespeople WHERE sname = 'Motika');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.00 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
D3_96509_Aditya> select * from orders o WHERE o.snum
    ->  = ANY (select snum FROM customers WHERE city = 'London');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.00 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
D3_96509_Aditya> select * from orders  WHERE amt > ANY (select amt from orders WHERE cnum =
 ANY (select cnum from customers WHERE city= 'London'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.00 sec)

D3_96509_Aditya>
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres’. (Hint: find all customers of serres salesperson).

```
D3_96509_Aditya> select * from customers WHERE rating >=
    -> ANY (select rating from customers WHERE snum = (select snum from salespeople WHERE s
name='Serres'));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)

D3_96509_Aditya>
```

11. Select customers who have a greater rating than any other customer in Rome.

```
D3_96509_Aditya> select * from customers WHERE rating > ANY (select rating from customers W
HERE cnum = ANY (select cnum from customers WHERE city= 'Rome'));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’.

```
D3_96509_Aditya> select * from orders WHERE amt >ANY (select amt from orders WHERE odate =
ANY (select odate from orders WHERE odate = '1990-10-04'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)

D3_96509_Aditya>
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
D3_96509_Aditya> select * from orders WHERE amt < ANY (select amt from orders WHERE cnum =
ANY (select cnum from customers WHERE city = 'San Jose'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average

amount from the

```
D3_96509_Aditya> select c.cname,o.odate,o.amt FROM customers c
    -> INNER JOIN orders o ON o.cnum = c.cnum
    -> WHERE o.amt >= (select AVG(amt) FROM orders);
+---------+------------+---------+
| cname   | odate      | amt     |
+---------+------------+---------+
| Liu     | 1990-10-03 | 5160.45 |
| Clemens | 1990-10-04 | 9891.88 |
| Clemens | 1990-10-04 | 4723.00 |
+---------+------------+---------+
3 rows in set (0.00 sec)
```

orders table.

13. Select those customers whose rating are higher than every customer in London.

```
D3_96509_Aditya> select * from customers WHERE rating >ALL (select rating from customers WH
ERE city = 'London');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)

D3_96509_Aditya> |
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
D3_96509_Aditya> SELECT snum, sname, city
    -> FROM salespeople
    -> WHERE city <> ALL (
    ->     SELECT city
    ->     FROM customers
    ->     WHERE customers.snum = salespeople.snum
    -> );
+------+---------+-----------+
| snum | sname   | city      |
+------+---------+-----------+
| 1004 | Motika  | London    |
| 1007 | Rifkin  | Barcelona |
| 1003 | Axelrod | New York  |
+------+---------+-----------+
3 rows in set (0.00 sec)

D3_96509_Aditya> |
```

9. Find names and numbers of all salesperson who have more than one customer.

```
D3_96509_Aditya> select * from salespeople s
    -> WHERE s.snum NOT IN (select c.snum FROM customers c
    ->                      WHERE c.city = s.city);
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.00 sec)

D3_96509_Aditya>
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
D3_96509_Aditya> SELECT
    ->     snum,
    ->     SUM(amt) AS total_amount
    -> FROM orders
    -> GROUP BY snum
    -> HAVING SUM(amt) > (SELECT MAX(amt) FROM orders);
+------+--------------+
| snum | total_amount |
+------+--------------+
| 1001 |     15382.07 |
+------+--------------+
1 row in set (0.00 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the

```
D3_96509_Aditya> SELECT
    ->     snum,
    ->     SUM(amt) AS total_amount
    -> FROM orders
    -> GROUP BY snum
    -> HAVING SUM(amt) > (SELECT MAX(amt) FROM orders);
+------+--------------+
| snum | total_amount |
+------+--------------+
| 1001 |     15382.07 |
+------+--------------+
1 row in set (0.00 sec)
```

table.